

Basic Principles of Medicine 1
Module : Cardio, Pulmonary, Renal
Lecture: CPR 32

Hemoglobin and Gas Transport

Dr. Shellon Thomas

sthoma10@sgu.edu

Dr. Sharmila Upadhya, MBBS, MD

shupadhya@sgu.edu

St Georges University ©

Copyright

All year 1 courses materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed or published in any form, printed, electronic or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

View only files, such as lecture recordings, are explicitly excluded from download and creating copies of these recordings by students and other users are strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St George's University. Users of this document identifying potential violations of these regulations are asked to bring their concern to the attention of the author.

Lecture: Hemoglobin and gas transport

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0286	Analyze how hemoglobin can regulate its oxygen affinity to pick up oxygen in the lungs, and deliver oxygen to the tissues
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0287	Discuss the structure-function relationships in disorders associated with hemoglobin.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0288	Discuss fetal hemoglobin and explain why it has a higher affinity for oxygen compared to HbA
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0289	Distinguish between hemoglobin (Hb) and myoglobin structures.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0290	Describe the conformational change in the Hb molecule following oxygen binding.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0291	Outline how hemoglobin participates in oxygen and carbon dioxide transport in blood
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0292	Discuss how carbon monoxide (CO) binds to Hb and how this interferes with normal oxygen binding.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0293	Discuss the significance of the sigmoidal oxygen binding curve of Hb, and how this is produced by cooperative binding of O ₂ to Hb.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0294	Describe the function of 2,3-BPG in regulating oxygen binding to Hb. Analyze the role of 2,3 BPG in adaptation to high altitude, and its role in oxygen binding to fetal hemoglobin, and in blood transfusion.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0295	Explain the Bohr effect and its significance in regulating oxygen release from oxyHb (H ⁺ , CO ₂).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0296	Compare and contrast the structure of HbA, HbA ₂ , HbA _{1C} and HbF.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0297	Describe methemoglobin and its impact on the structure and function of hemoglobin

Recommended Reading

- Lippincott's Biochemistry 9th edition Chapter 3: Globular Proteins
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- [Globular Proteins | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- **Questions from the Lippincott's 9th edition book - Chapter 3**
 - Questions: 3.1-3.7
 - Concept maps on Fig.3.24 and Fig.3.25
- Small group discussion case is pertinent
- **Sickle cell anemia in Lippincott's Biochemistry Book**
 - This topic will be covered in Hemoglobinopathies lecture/ FTM module

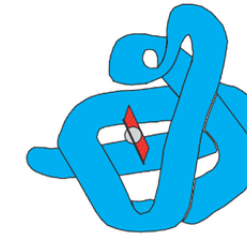
Oxygen binding to hemoglobin:

<http://www.youtube.com/watch?v=91c04OoTEJs>

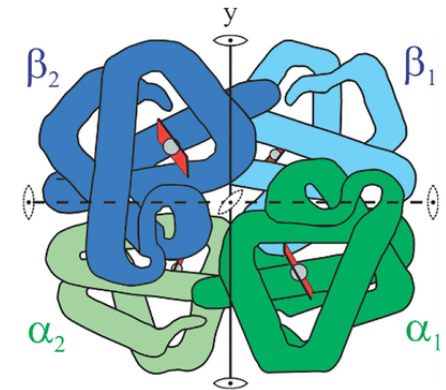


Hemoglobin and myoglobin

- Many similarities in primary structure
- Overall fold same for globin monomers: *globin fold*
- Key amino acid residues **conserved**
- **Myoglobin: Single polypeptide chain (Monomer)**
 - One Heme group with Fe²⁺
 - Binds one O₂ molecule
- **Hemoglobin: Four polypeptide chains (Tetramer)**
 - Four Heme groups with Fe²⁺
 - Binds 4 O₂ molecules

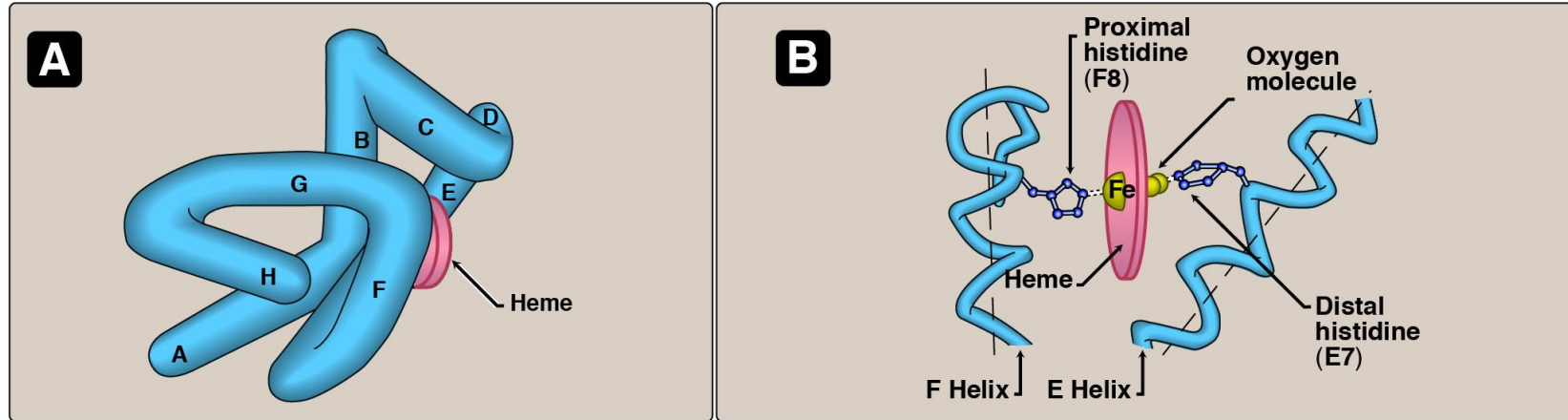


myoglobin



hemoglobin

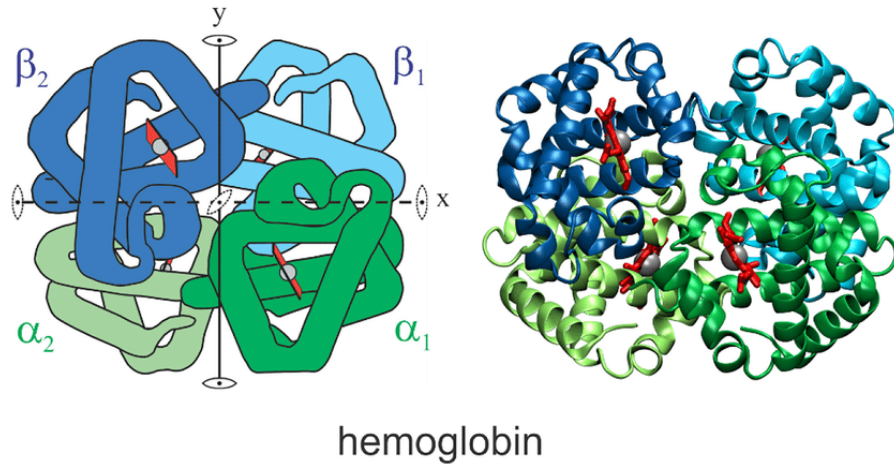
Myoglobin structure



Myoglobin with eight alpha helices (A-H)

Myoglobin's oxygen binding site

- Globin protein monomer mostly composed of **α -helix (labeled A – H helices)**
- A few bends or turns
- Globin structure is devoid of β -sheet (unusual)
- Heme group with Fe²⁺ (Oxygen binding site): Located in '**EF**' pocket
- Allows reversible binding of O₂

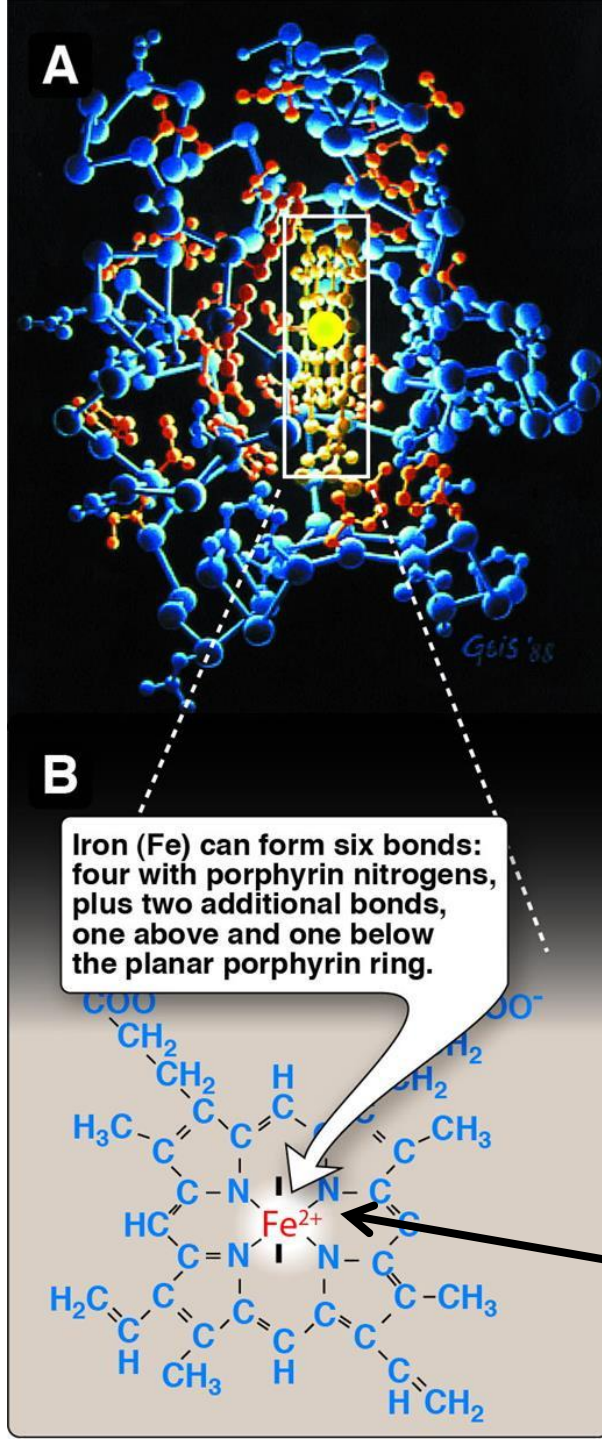


Hemoglobin tetramer

- $\alpha_2\beta_2$ tetramer
- Hemoglobin A (HbA) is a pair of identical $\alpha\beta$ dimers ($\alpha_2\beta_2$)
- Non-covalent interactions between subunits
 - Hydrophobic
 - Ionic bonds
 - Hydrogen bonds
- Four Heme groups
- Four globin monomers means 4 O_2 molecules may be carried on Hb
- 1 g Hb = 1.34 mL of O_2
- **15g Hb carries about 20.4 mL of O_2**

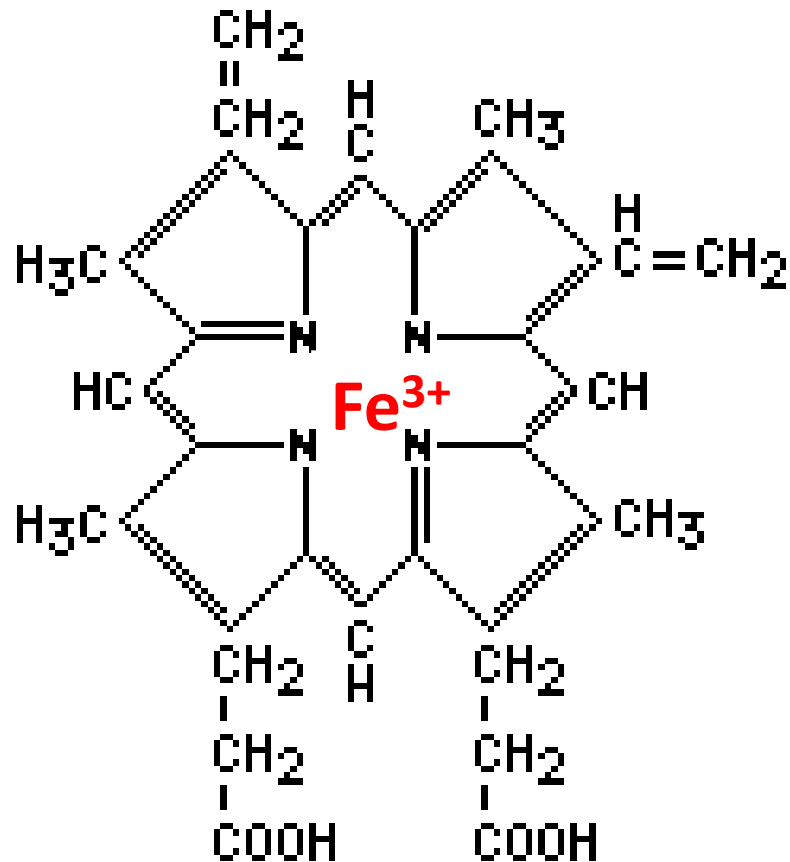
Oxygen binding site

- Fe^{2+} (Ferrous ion) may form **6 bonds**
 - Four bonds to nitrogen atoms in the porphyrin ring
 - One bond links to Histidine of globin protein
 - One bond reversibly binds oxygen
- Iron (Fe^{2+}) is buried deep inside the globin protein
 - Ensures that oxygen is released as O_2
 - Porphyrin ring (with iron) located in '**EF**' pocket



Methemoglobin

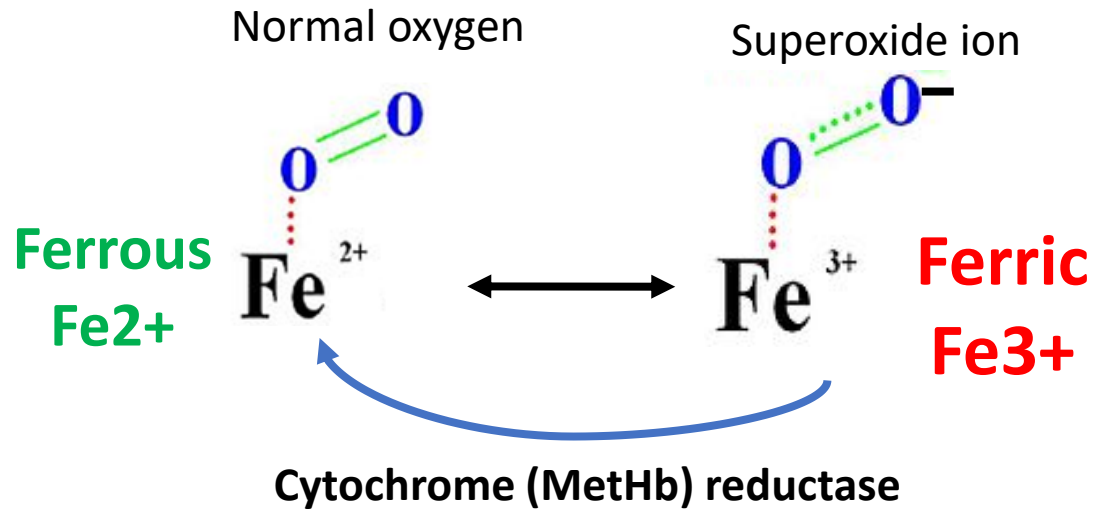
- Methemoglobin (MetHb) contains **Fe³⁺ (ferric ion)** in the porphyrin ring



Ferric Heme

- Fe²⁺ (gets oxidized) → **Fe³⁺**
 - Occurs when superoxide (O₂⁻¹) is released instead of molecular oxygen (O₂)
- Ferric ion is the **oxidized version** of the iron atom
 - **Ferric ion does not bind to O₂**
- Remaining 3 O₂ sites do not release the bound oxygen
 - Same argument as for carbon monoxide
 - *We will see later in this session*
- Therefore, **MetHb does not function as an O₂ carrier**

Methemoglobin



**Ferric ion does
not bind to O₂**



Martin Fugate and family are shown in this undated, colorized black and white photo

<https://abcnews.go.com/Health/blue-skinned-people-kentucky-reveal-todays-genetic-lesson/story?id=15759819>

<http://www.indiana.edu/~oso/lessons/Blues/TheBlues.htm>

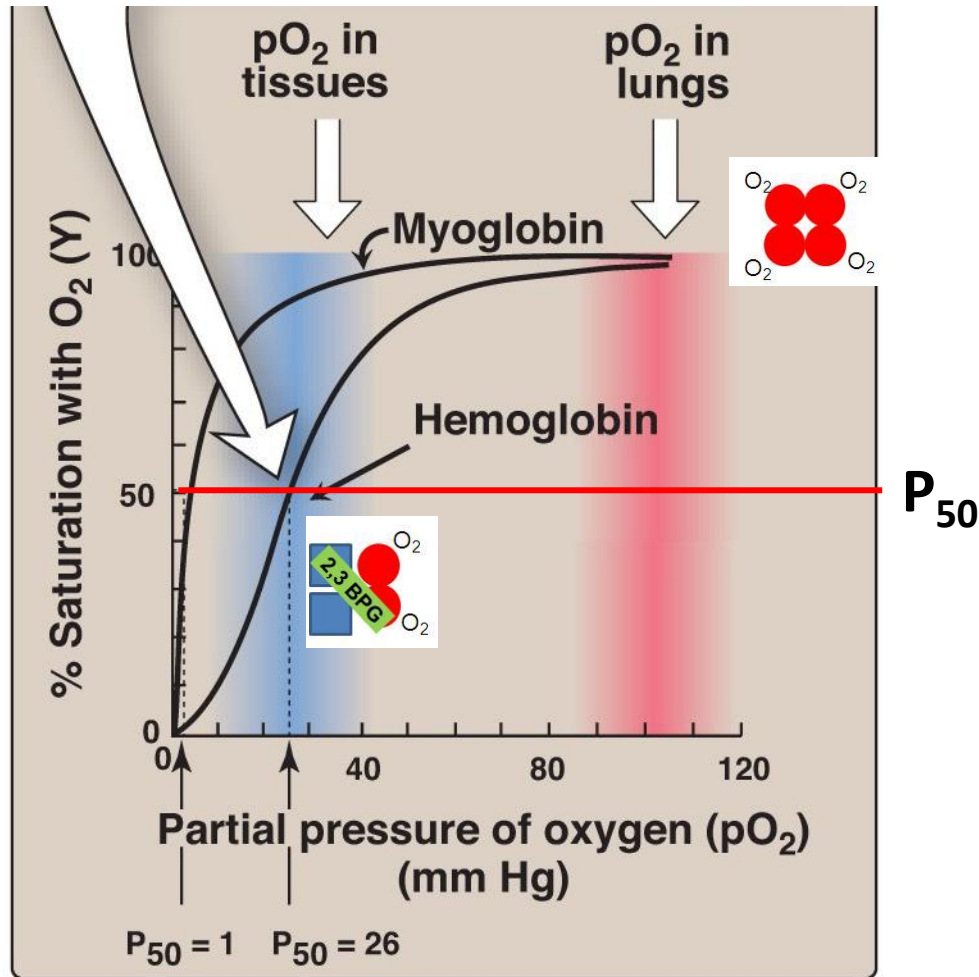


**Treatment: Methylene blue;
Vitamin C**

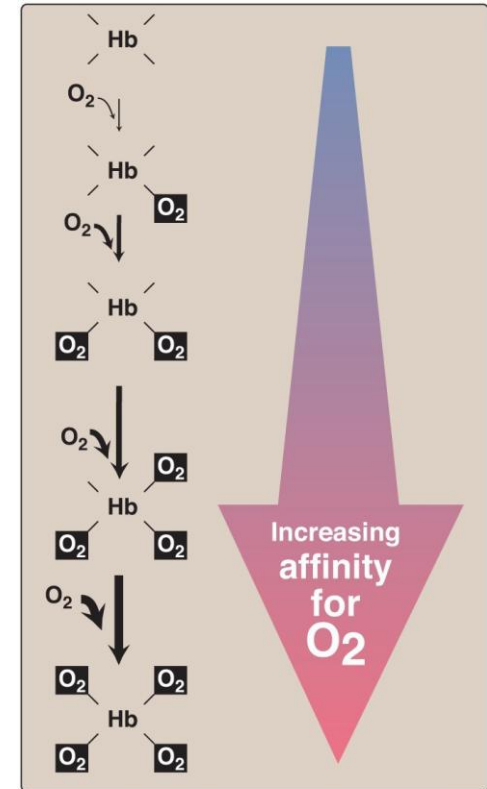
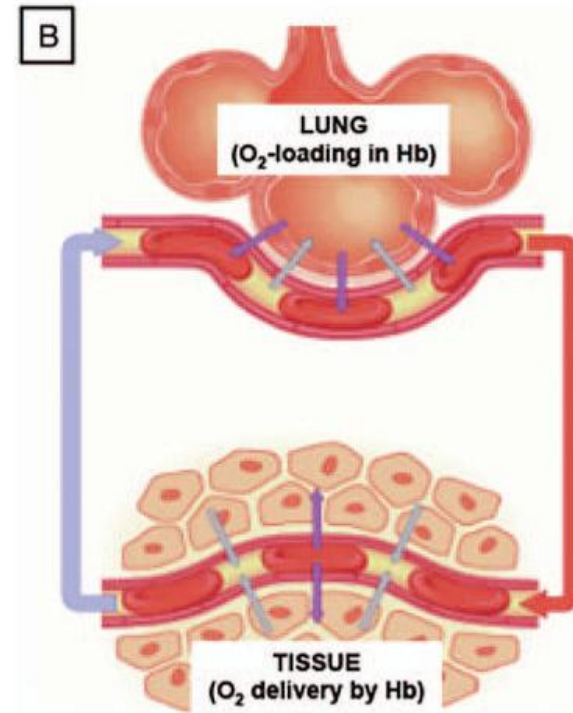
**Excessive
nitrites/nitrates:
Oxidant**



Myoglobin has a higher affinity for O_2 than Hb



Copyright © 2014 Wolters Kluwer Health | Lippincott Williams & Wilkins



Copyright © 2014 Wolters Kluwer Health | Lippincott Williams & Wilkins

Oxygen dissociation curve for myoglobin is hyperbolic

Hb binds oxygen with cooperativity, so its dissociation curve is sigmoidal

Cooperativity: Binding of one O_2 molecule promotes binding of another O_2 molecule

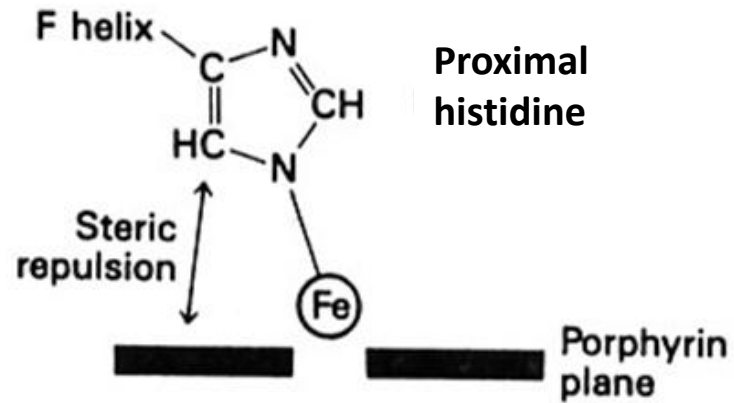
Comparison of Hb & myoglobin

	<u>Hemoglobin</u>	<u>Myoglobin</u>
★ Function	O ₂ transport	O ₂ storage
Location	Erythrocytes	Muscle tissue
Tissue O₂ affinity	Low	High
Lung O₂ affinity	High	-----
O₂ affinity change with pO₂	Yes	No
Allosteric regulation	Yes	No
★ Quaternary structure	Tetramer	Monomer (not quaternary)
★ Number of O₂ bound	Four	One
★ Oxygen dissociation curve	Sigmoidal (Cooperativity)	Hyperbolic

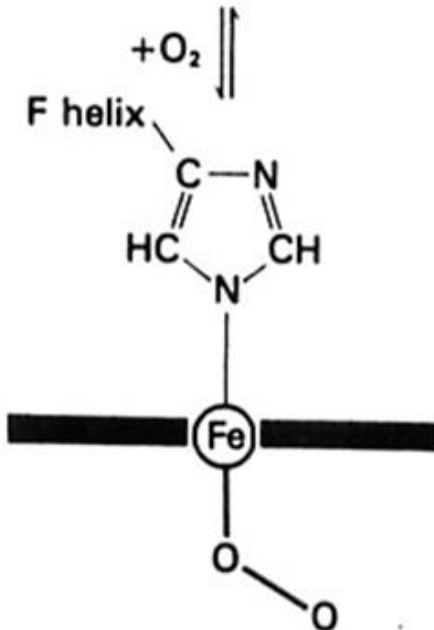
★ **Myoglobin stores oxygen in the muscle, and releases oxygen to be used during muscle contraction**

O₂ binding to heme leads to conformational change in hemoglobin

Globin chain

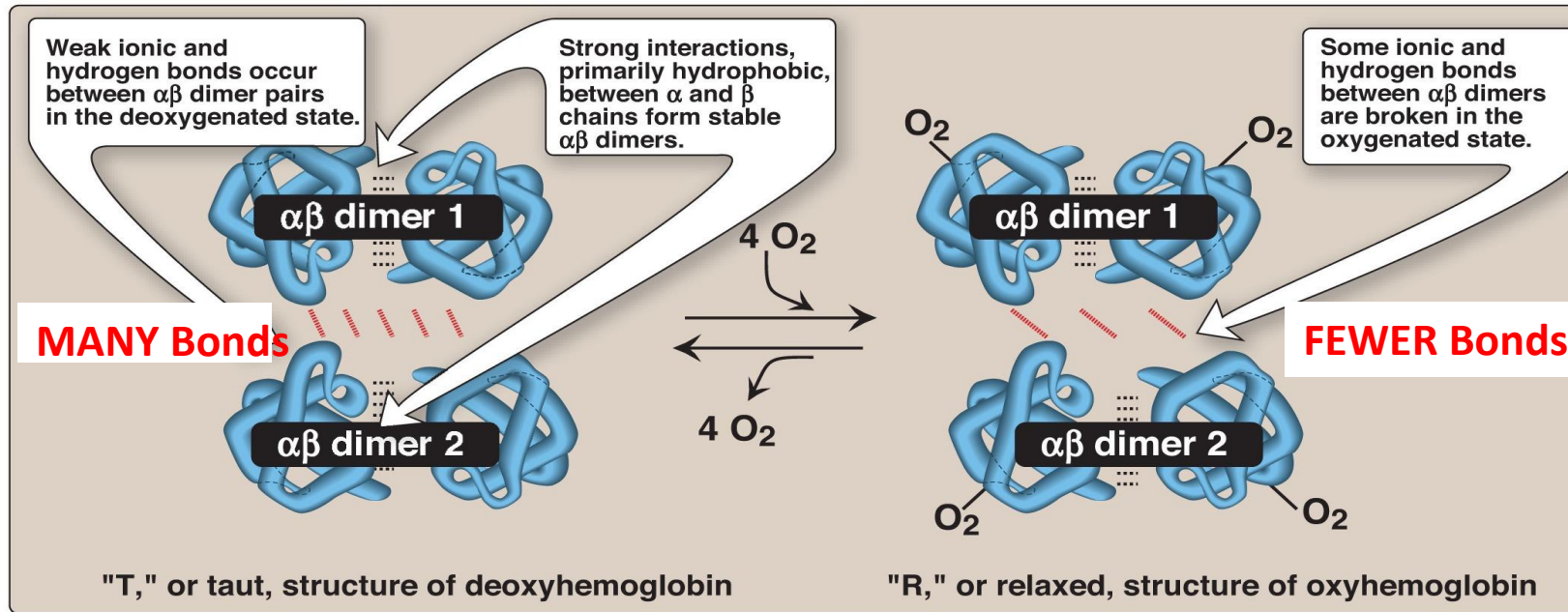


Globin chain



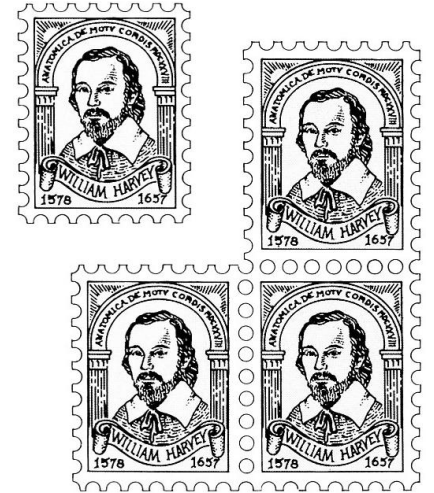
- Normally Fe²⁺ ion is slightly out of plane of heme ring
- Fe²⁺ atom (ferrous) moves into the plane of heme when oxygenated
- Proximal histidine and its associated globin chain pulled along with iron ion (tugging)
 - Conformational change is transmitted to the other globin monomers in Hb
- Explains conformational change associated with oxygen binding
 - Explains cooperative oxygen binding

'R' and 'T' states of hemoglobin and Cooperativity



Low oxygen affinity

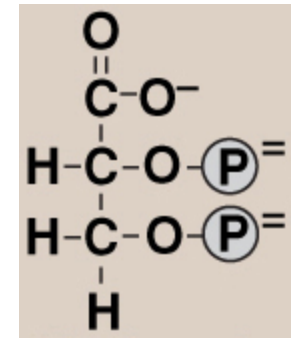
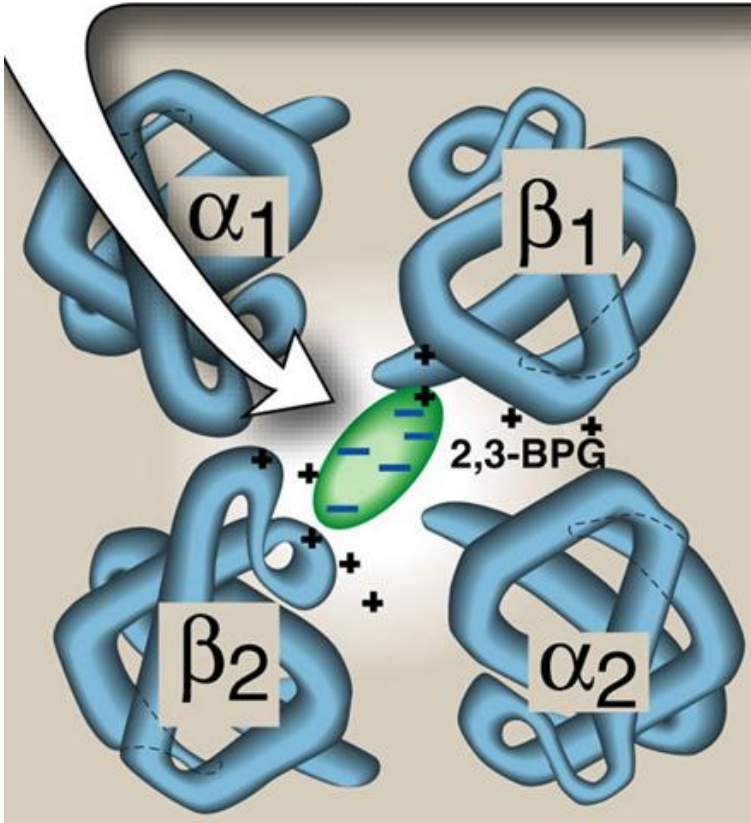
**As Hb binds oxygen,
its oxygen affinity
increases**



- Two perforated edges torn to remove the first stamp
 - First O_2 binds with least affinity
- Successive stamp removal easier as more stamps are removed
 - Medium affinity O_2 binding
- Last stamp is free without tearing any edges
 - Last O_2 binds with highest affinity

2,3-BPG promotes taut “T” state

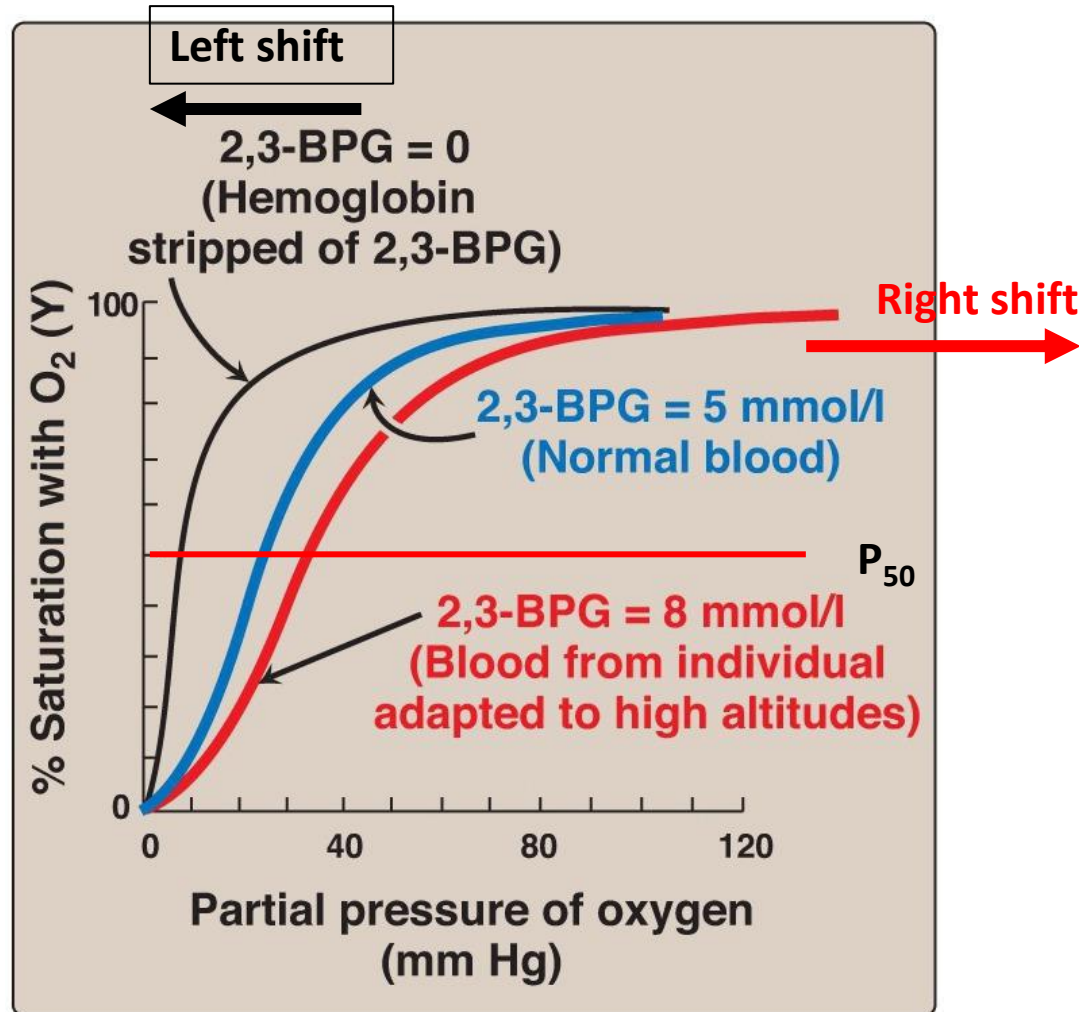
A single molecule of 2,3-BPG binds to a positively charged cavity formed by the β -chains of deoxyhemoglobin.



2,3-Bisphospho Glycerate

- 2,3-BPG (or 2,3-DPG) is negatively charged
 - Phosphate groups (PO_4^{-3})
 - Carboxylate groups (COO^-)
- By-product of **glycolysis**
- **Facilitates Hb to assume deoxyHb structure (taut ‘T’ form)**
 - Promotes unloading O_2 **in tissues**
- **Decreases O_2 affinity for Hb**

2,3-BPG decreases Hb oxygen affinity

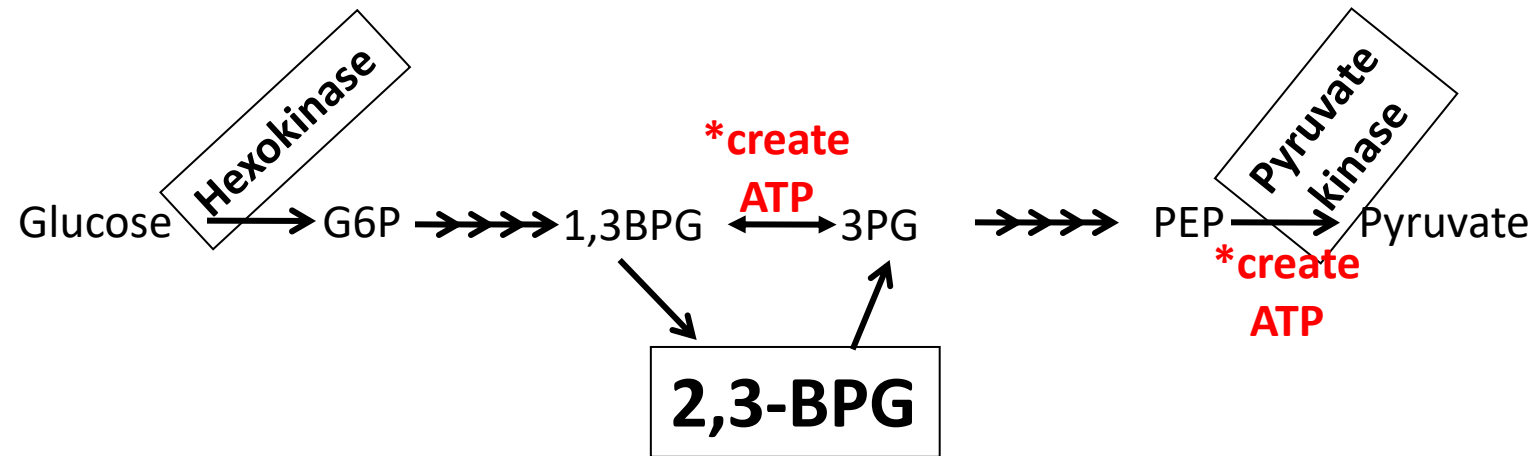


Copyright © 2014 Wolters Kluwer Health | Lippincott Williams & Wilkins

- Higher levels of 2,3-BPG promote oxygen release to the tissues
- Decreases the affinity of Hb for oxygen
- 2,3-BPG levels are increased at high altitude (adaptation)
 - Facilitates oxygen unloading in the tissues

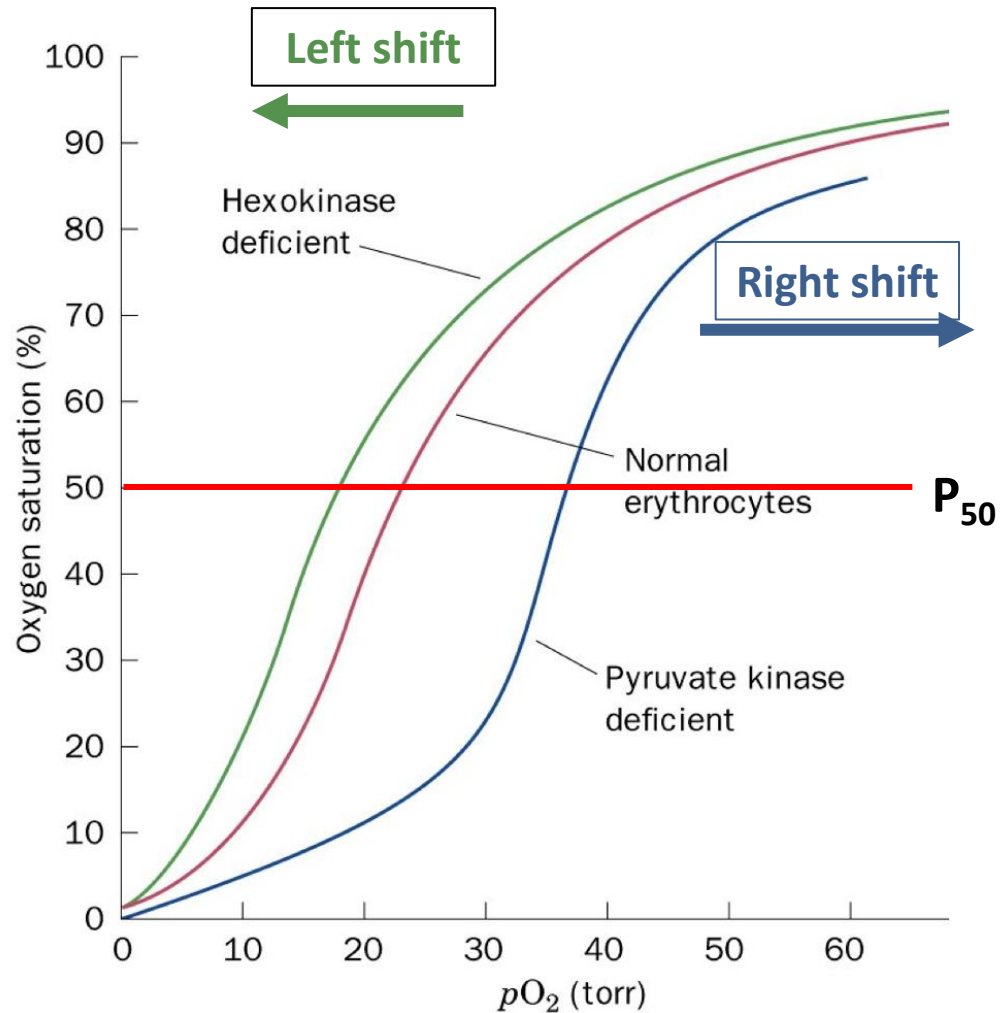
2,3-BPG promotes oxygen release from hemoglobin

Partial deficiencies in glycolysis may affect O₂ binding affinity of hemoglobin



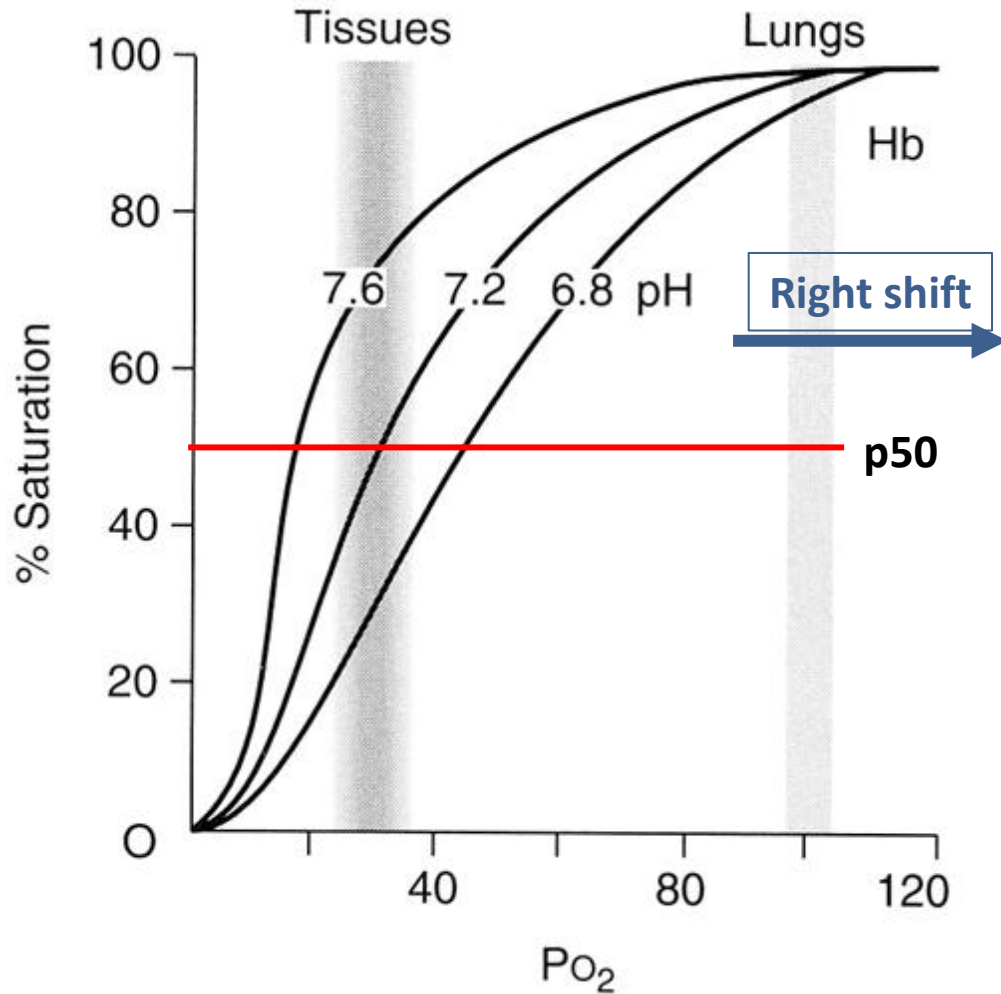
Compare effects of hexokinase (HK) deficiency to pyruvate kinase (PK) deficiency in the glycolysis pathway

Pyruvate kinase Vs Hexokinase deficiency



- **Hexokinase deficiency**
 - Reduced 2,3-BPG levels
 - Left shift on binding curve
 - Increased O_2 affinity
- **Pyruvate kinase deficiency**
 - Elevated 2,3-BPG levels
 - Right shift on binding curve
 - Decreased O_2 affinity

H⁺ ions promote oxygen release from Hb



- As pH decreases, Hb O₂ affinity decreases
- **Bohr effect**
- Logic:
 - Aerobic tissues form CO₂ during metabolism
 - Dissolved CO₂ becomes an acid when it reacts with H₂O
 - Anaerobic tissue also produces lactic acid
- CO₂ and H⁺ ions signal a need for more O₂

H⁺ and CO₂ have no effect on O₂ binding to myoglobin

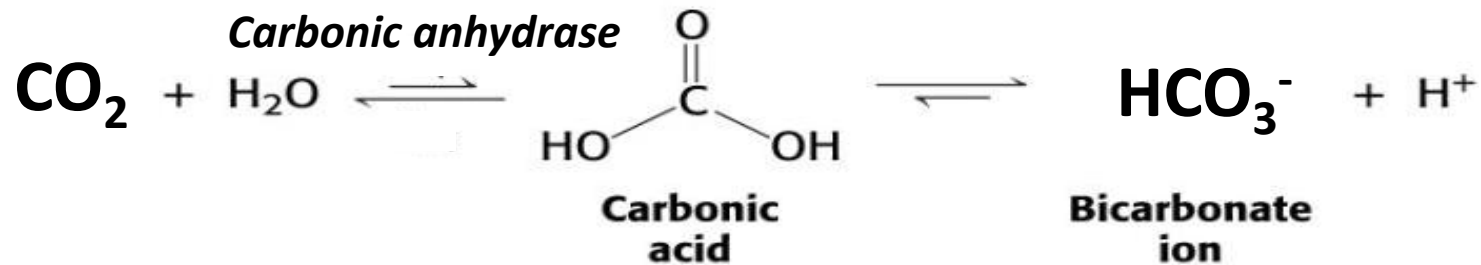
Carbon dioxide transport

- Hemoglobin (**Histidine residues**) mops up the protons generated during CO₂ transport (volatile acid)
- **Hb responsible for significant proportion of acid buffering during CO₂ transport**
- **CO₂ is transported in blood as**
 - HCO₃⁻
 - CarbaminoHb: CO₂ binding to N-terminal of globin chains
 - Dissolved CO₂ (PCO₂)

CO₂ transport in the blood: <https://www.khanacademy.org/science/health-and-medicine/hematologic%20system/v/hemoglobin-moves-o2-and-co2>

Carbon dioxide transport

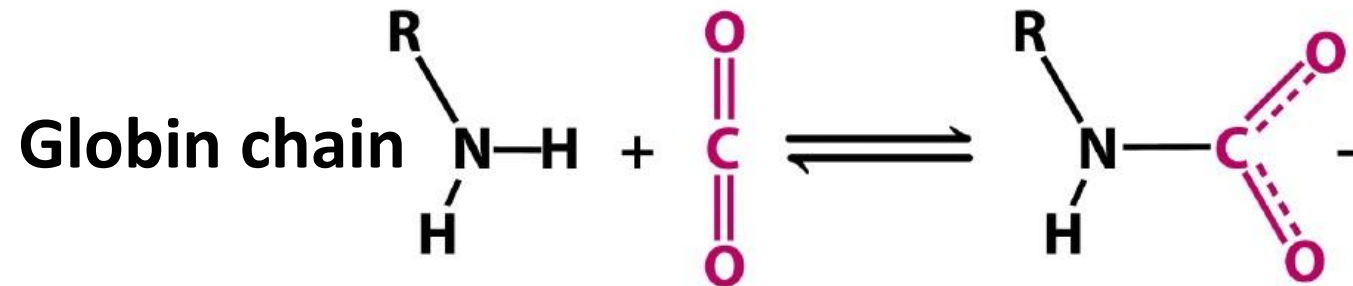
1. Bicarbonate (HCO_3^-)



H^+ ions bind to **Histidine** residues of Hb, forms positive charges

Formation of additional salt bridges (bonds)

Stabilizes taut form of Hb; promotes O_2 delivery to tissues

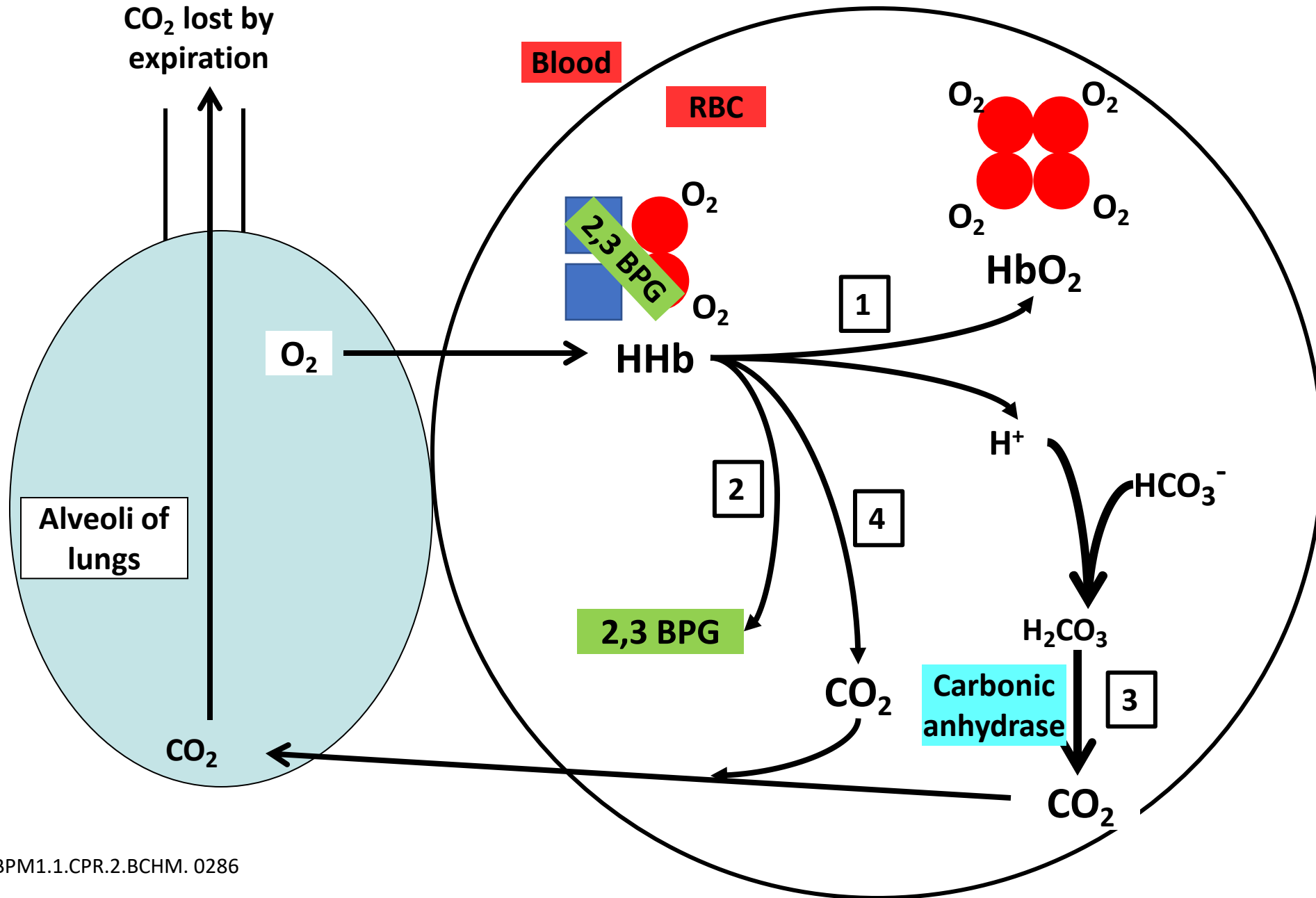


2. Carbaminohemoglobin

CO_2 binds to amino-terminus of globin-chains → **Carbamino** hemoglobin.

Allows formation of additional salt bridges (contributes to Bohr effect)

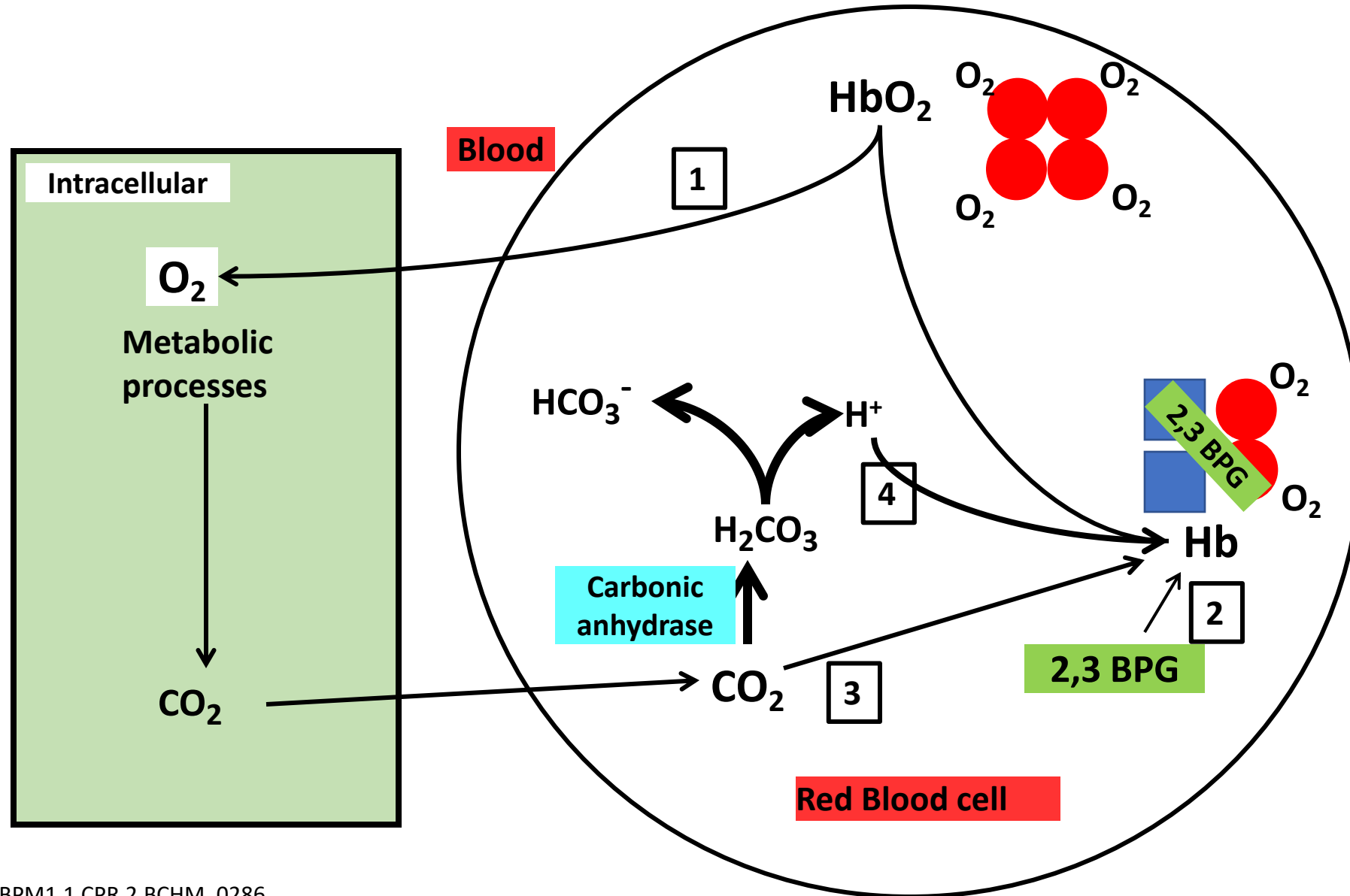
Events in the Lungs



Events in the Lungs

- In the **lungs**, CO_2 and H^+ leave Hb, which promotes release of 2,3-BPG, as O_2 binds
- High partial pressure of O_2 (PO_2) in the lungs overcomes the T-state, so O_2 binds to Hb (R state) (1)
- This forces out the 2,3-BPG (2)
- H^+ ions are released $\rightarrow$ H^+ associates with HCO_3^- to form carbonic acid which decomposes to form CO_2 (3)
- Also, CO_2 bound to the N-terminal end of globin chains (carbamino hemoglobin) is released (4)

Events in the tissues



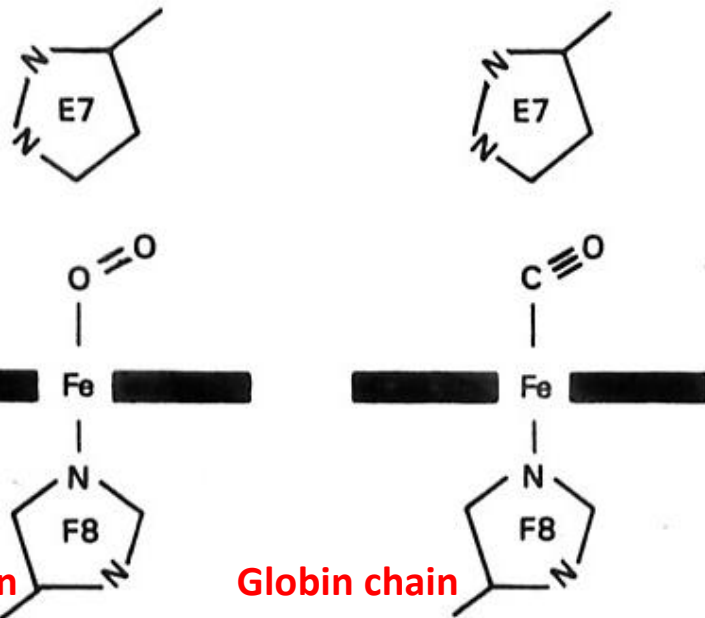
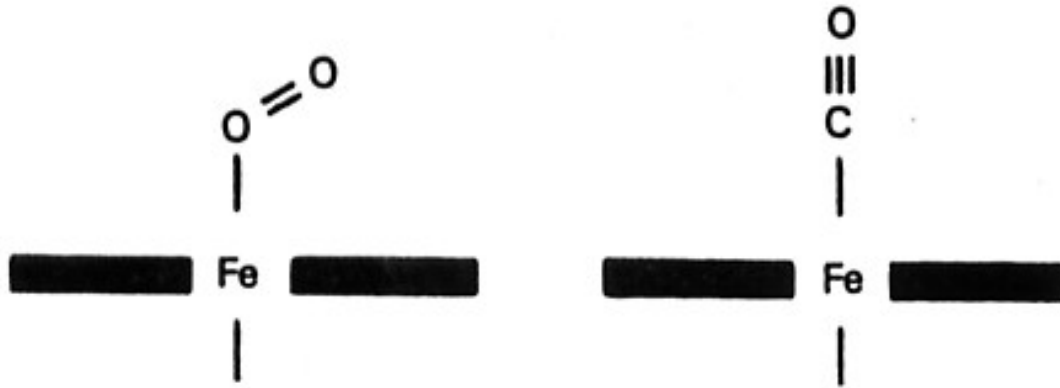
Events in the tissues

- Low $PO_2 \rightarrow O_2$ is liberated from Hb (1)
- CO_2 forms HCO_3^- and H^+ (by carbonic anhydrase); H^+ binds to Hb (4)
- CO_2 binds to N-terminal chains of Hb to form carbamino Hb (3)
- In the tissues, CO_2 and H^+ ions bind to Hb, which promotes binding of 2,3-BPG (2), and the delivery of $O_2 \rightarrow$ Favors T state
- H^+ ions, CO_2 and 2,3-BPG enhance oxygen delivery by decreasing O_2 affinity in the tissues. The “**T state**” is favored in the presence of H^+ ions, CO_2 and 2,3-BPG

Carbon monoxide

Free heme binds CO **25,000x**
more tightly than to O₂

Bonding angles for O₂ and CO to iron (ferrous) of heme



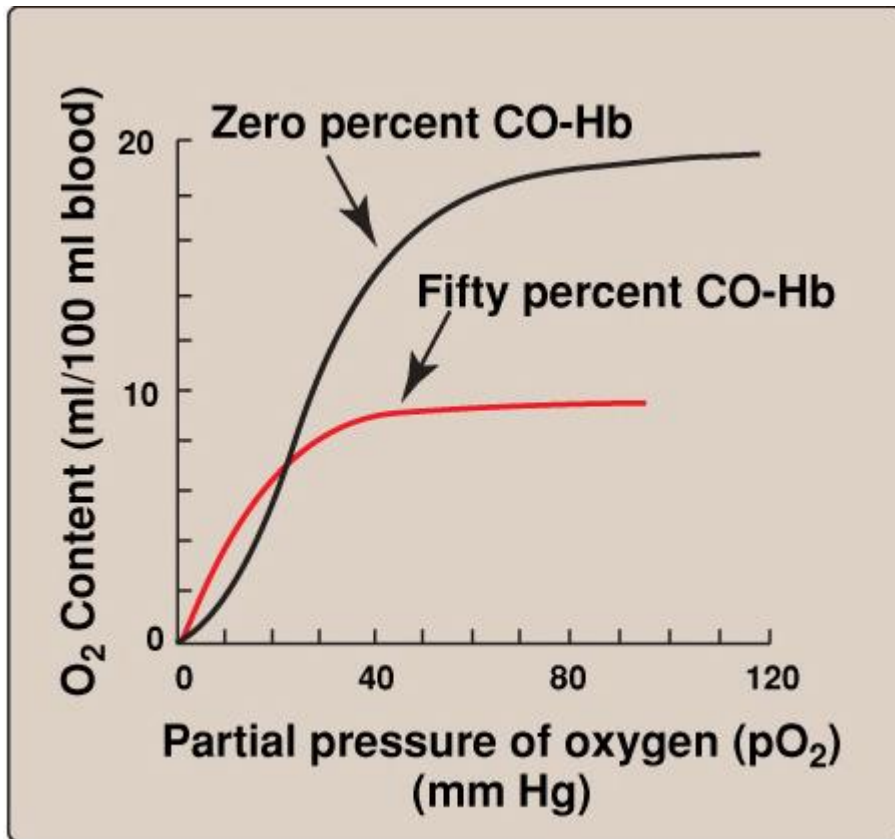
Globin chain

Globin chain

- Globin chain forces CO bond in a strained manner → weakens interaction
- CO binds to hemoglobin about **200x** stronger than oxygen (Bind to same site)
- CO binding to Hb: **stronger than O₂** and reversible

Angles for bonding of oxygen and carbon monoxide to the heme iron of myoglobin. The distal E7 histidine hinders bonding of CO at the preferred (180-degree) angle to the plane of the heme ring.

Carboxyhemoglobin



Copyright © 2008 Wolters Kluwer Health | Lippincott Williams & Wilkins

- Person who has 50% of Hb oxygen binding sites occupied by CO is near death (if not dead already)
- CO stabilizes '**R**' state of Hb → Decreases oxygen unloading by other subunits (Hyperbolic ODC)
- **Lower oxygen content; PaO₂ is normal**
- Compare to: Person who has anemia which reduces his Hb by 50% may still be alive.

Carboxyhemoglobin: Characteristic 'cherry red' appearance
ER physicians say: "cherry red is dead"

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0292

CARBON MONOXIDE
How to spot something that's colorless, odorless, and invisible?

Slight Headache	Severe Headache	Unconsciousness
Nausea	Confusion	Convulsions
Vomiting	Drowsiness	Cardio-respiratory Failure
Fatigue	Fast Heart Rate	Death

Stop it, before it stops you.

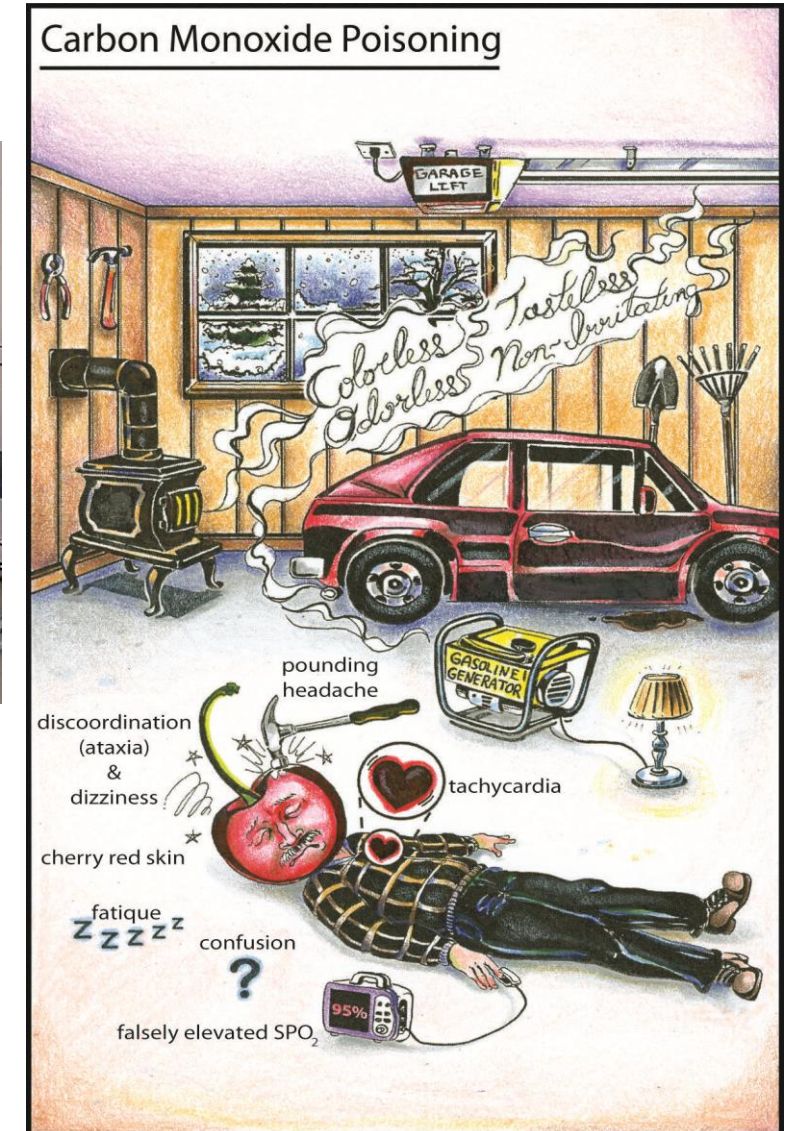
MICHIGANAUTOLAW
AUTO ACCIDENT ATTORNEYS

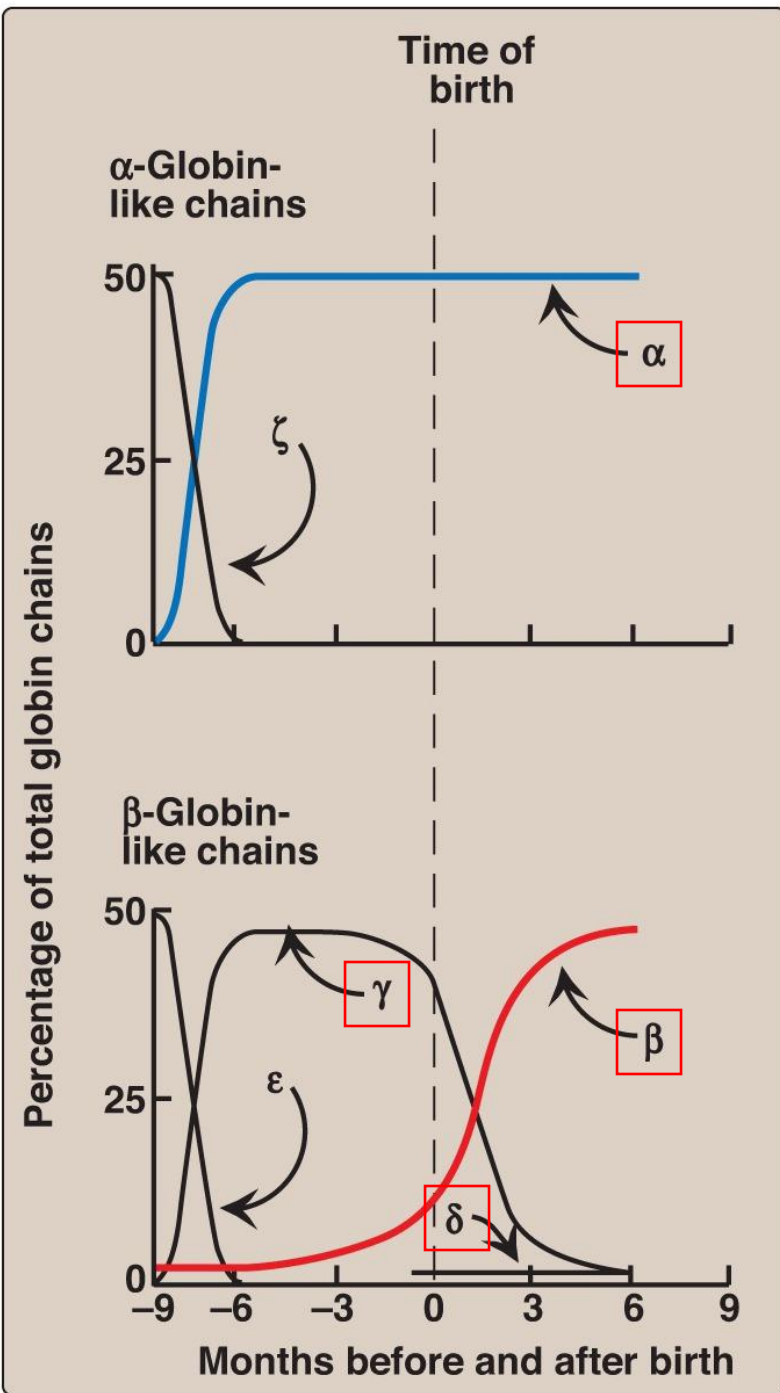
Carboxyhemoglobin: High oxygen affinity

- Management:
 - Remove toxic CO
 - Use 100% oxygen
 - Sometimes under elevated pressure
- Elevated oxygen pressure also has risks



Hyperbaric chamber





Keep track of α , β , γ , and δ :

- The α -chain is the same in all Hb's (after embryo)
- The γ -chain disappears soon after birth (HbF)
- The δ -chain is found in low amounts in adults (HbA₂)

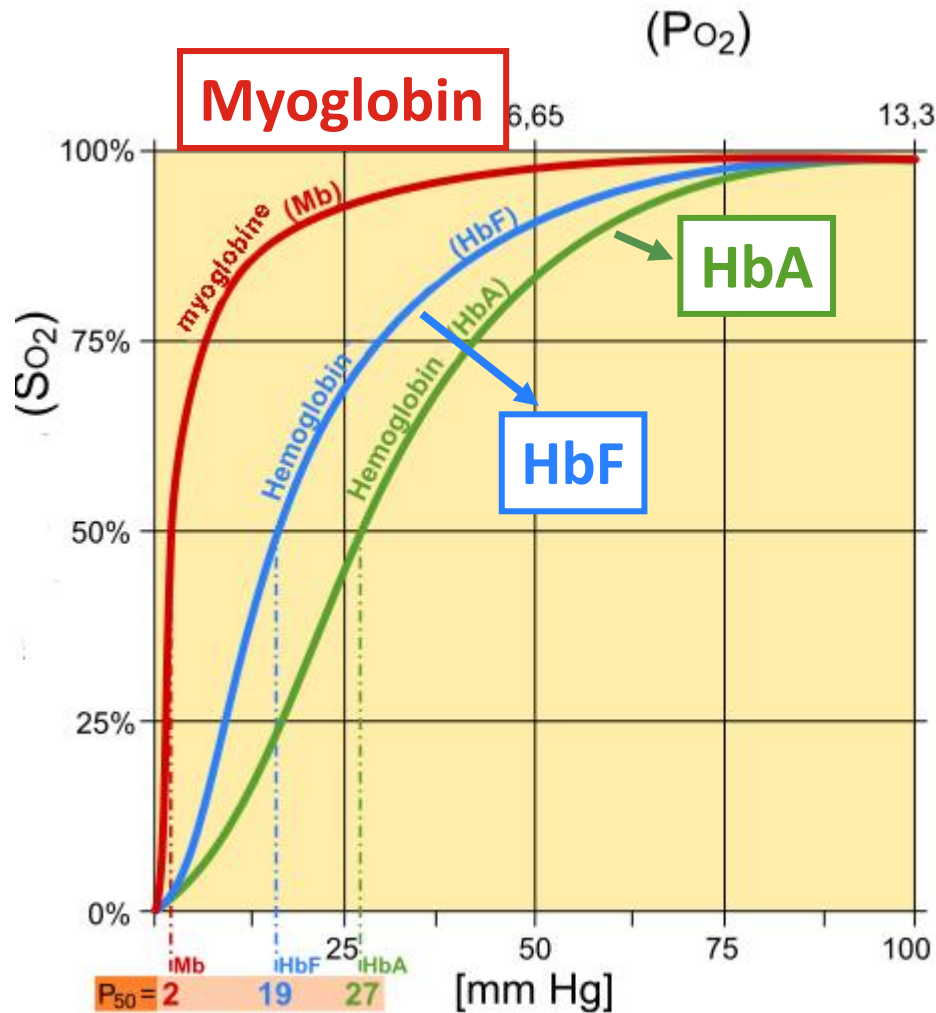
Form	Chain composition	Fraction of total hemoglobin
HbA	$\alpha_2\beta_2$	90%
HbF	$\alpha_2\gamma_2$	<2%
HbA ₂	$\alpha_2\delta_2$	2%–5%
HbA _{1c}	$\alpha_2\beta_2$ -glucose	3%–9%

Copyright © 2014 Wolters Kluwer Health | Lippincott Williams & Wilkins

Normal adult human hemoglobin types

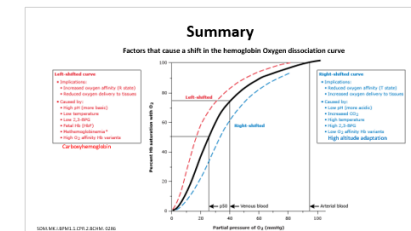
Note: α -chains are identical in all

Hemoglobin F binds 2,3-BPG less tightly → Slightly higher O₂ affinity



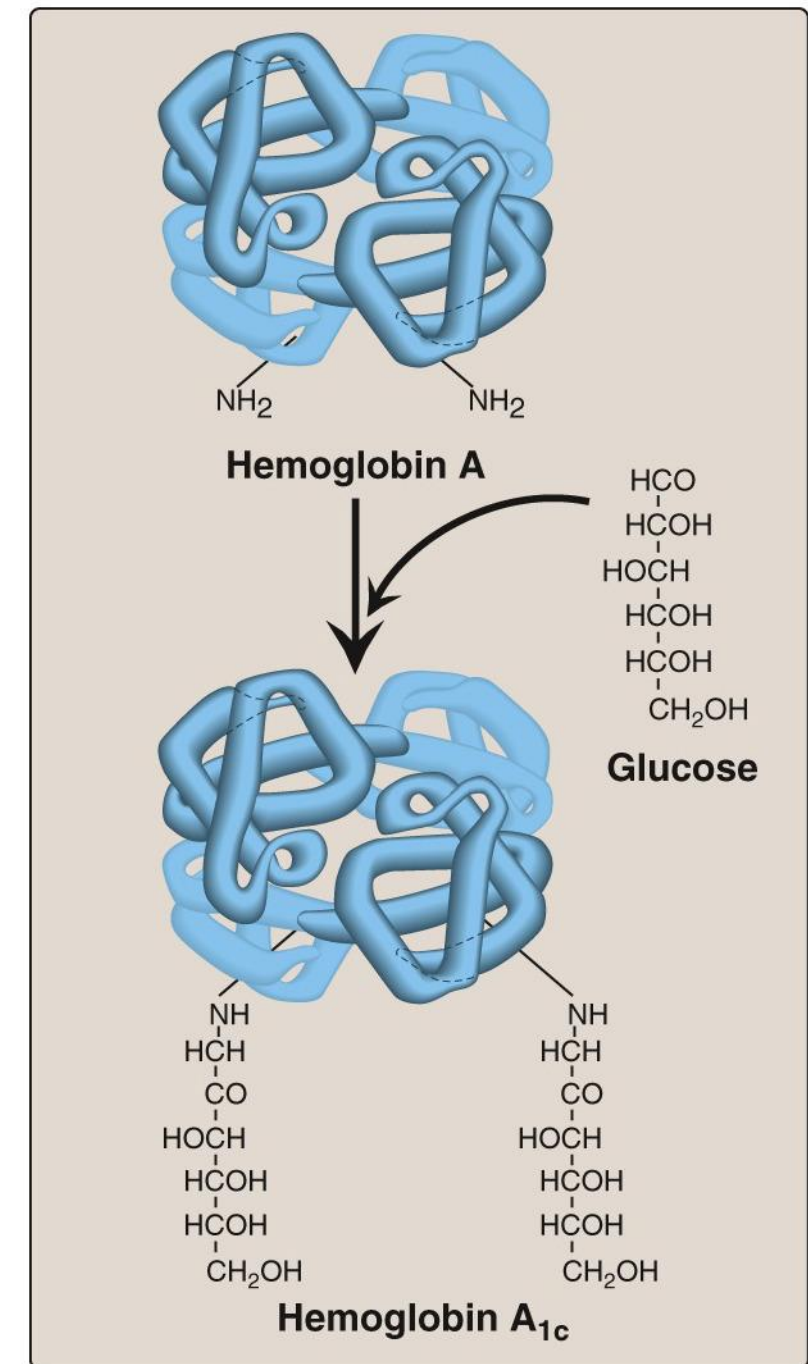
HbA $\alpha_2\beta_2$
HbF $\alpha_2\gamma_2$

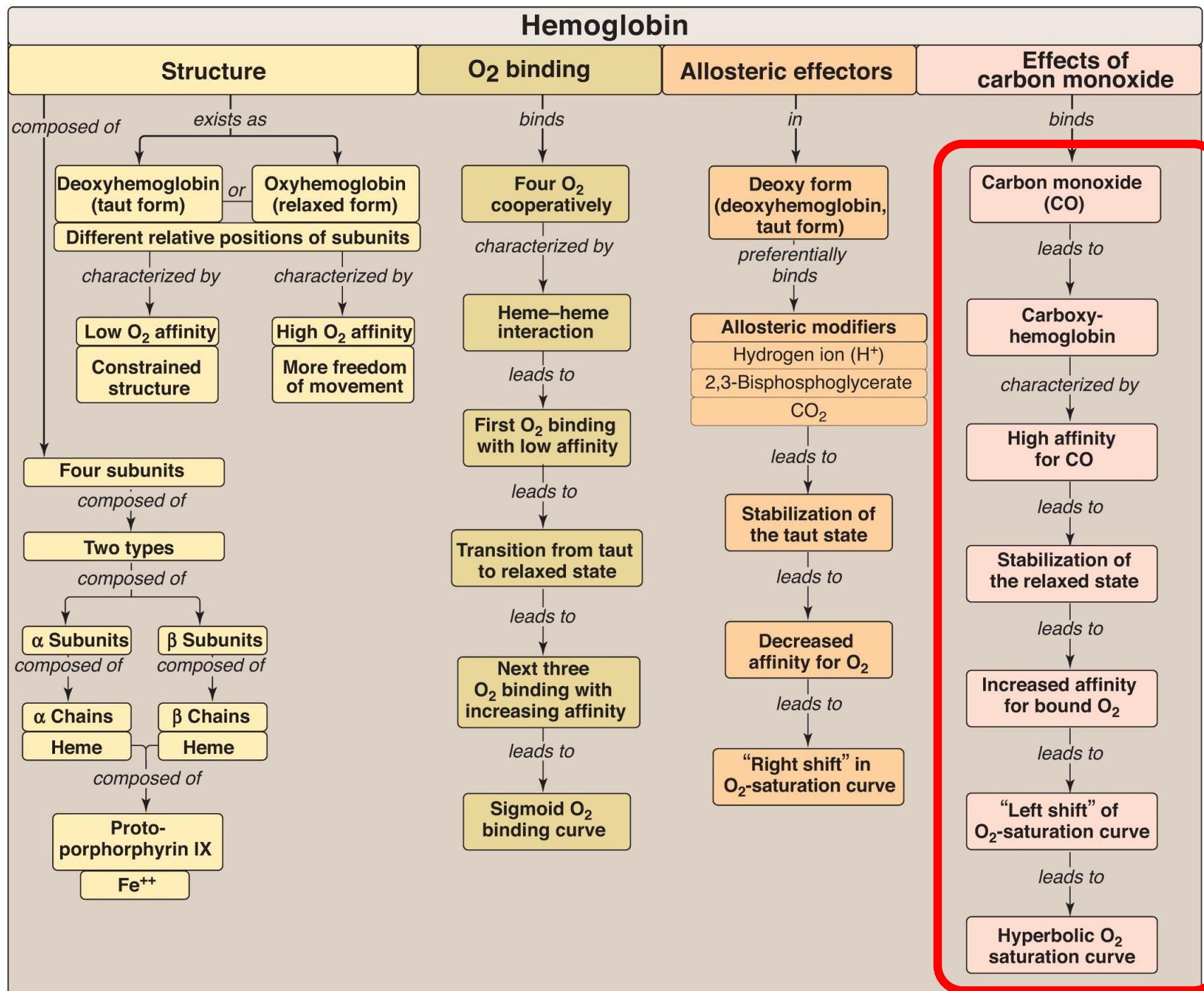
- Some adults express $\alpha_2\gamma_2$ at high levels
- Persistence of fetal Hb
 - Benign condition
 - If able to induce, may be a “cure” for β -thalassemia and for sickle cell disease



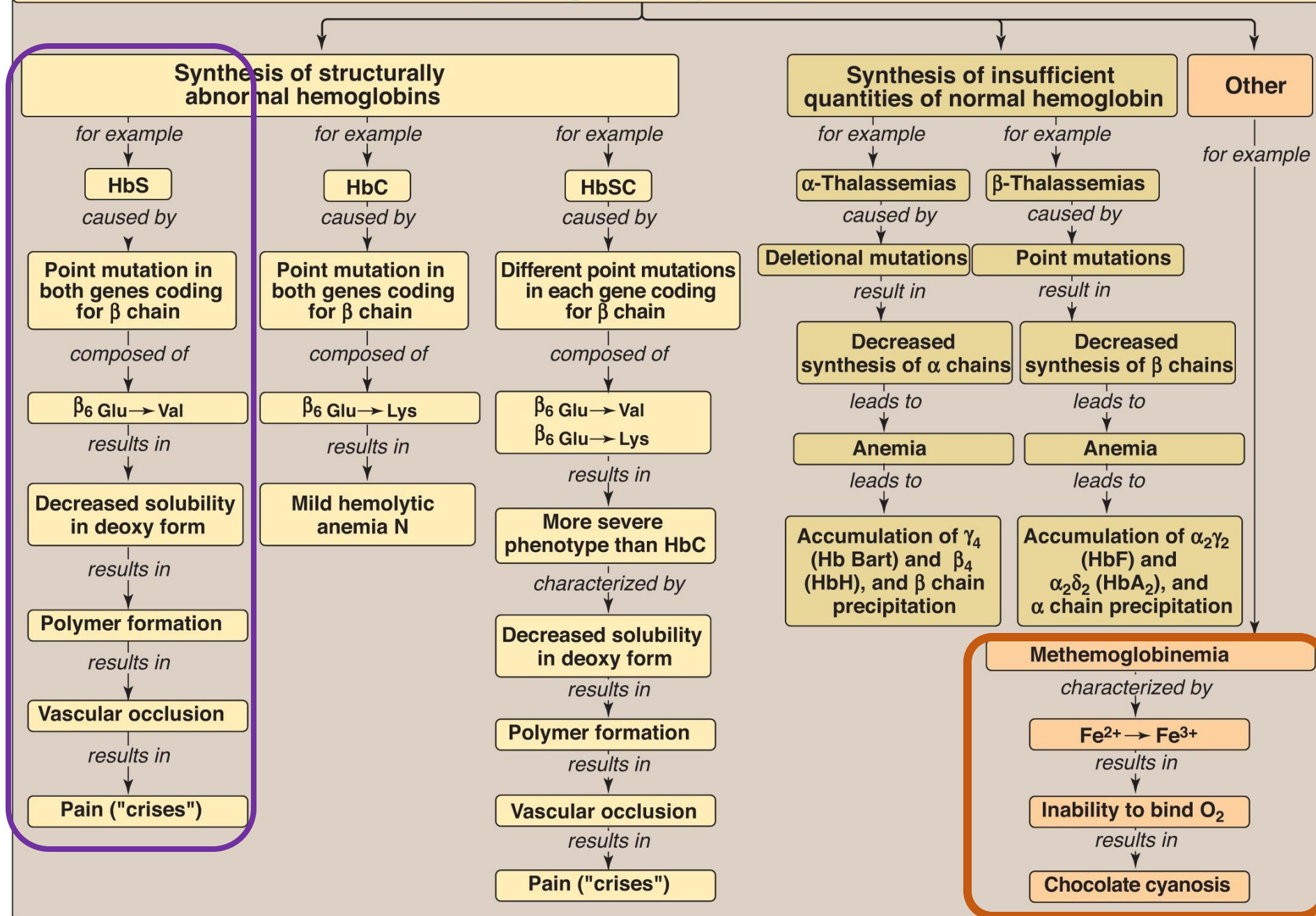
Glycated hemoglobin (HbA1c)

- N-terminus of β -globin may be **glycated**
 - Non-enzymatic, spontaneous reaction
 - **Hb** $\rightarrow$ non-enzymatic glycation $\rightarrow$ **HbA_{1c}**
 - **Depends on plasma glucose levels**
- **HbA_{1c}** formation has no effect on function of Hb
- HbA_{1c} levels: Plasma glucose levels over the previous three months (RBC lifespan)
 - Diagnosis and management of diabetes mellitus





Hemoglobinopathies

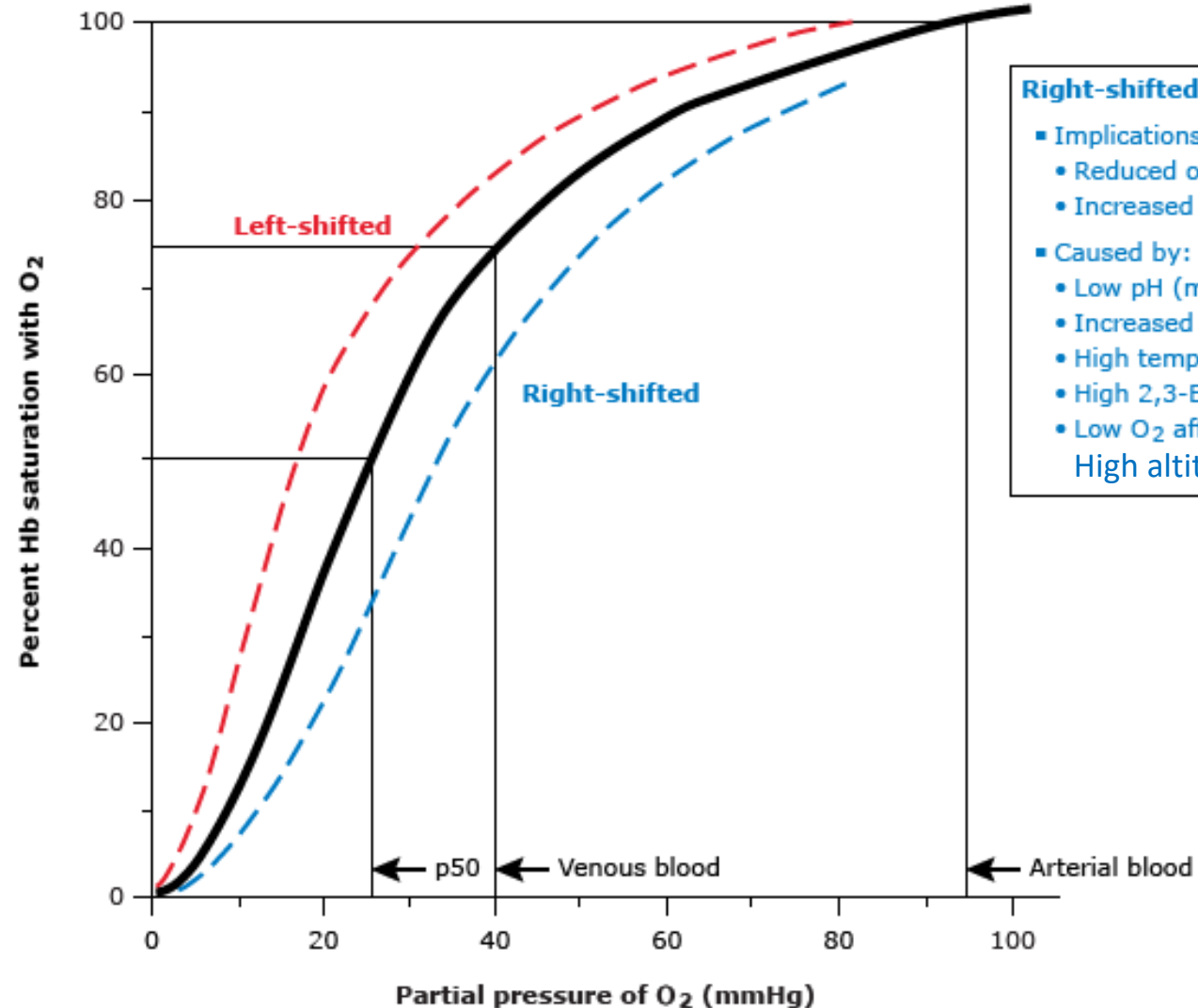


Summary

Factors that cause a shift in the hemoglobin Oxygen dissociation curve

- Left-shifted curve**
- Implications:
 - Increased oxygen affinity (R state)
 - Reduced oxygen delivery to tissues
 - Caused by:
 - High pH (more basic)
 - Low temperature
 - Low 2,3-BPG
 - Fetal Hb (HbF)
 - Methemoglobinemia*
 - High O₂ affinity Hb variants

Carboxyhemoglobin



- Right-shifted curve**
- Implications:
 - Reduced oxygen affinity (T state)
 - Increased oxygen delivery to tissues
 - Caused by:
 - Low pH (more acidic)
 - Increased CO₂
 - High temperature
 - High 2,3-BPG
 - Low O₂ affinity Hb variants
 - High altitude adaptation